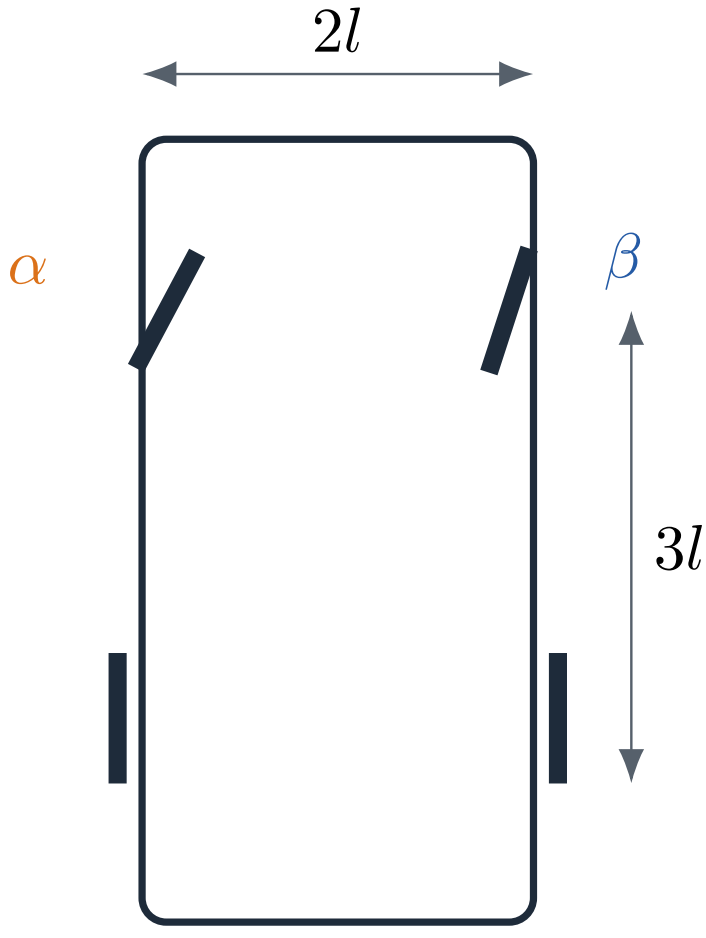
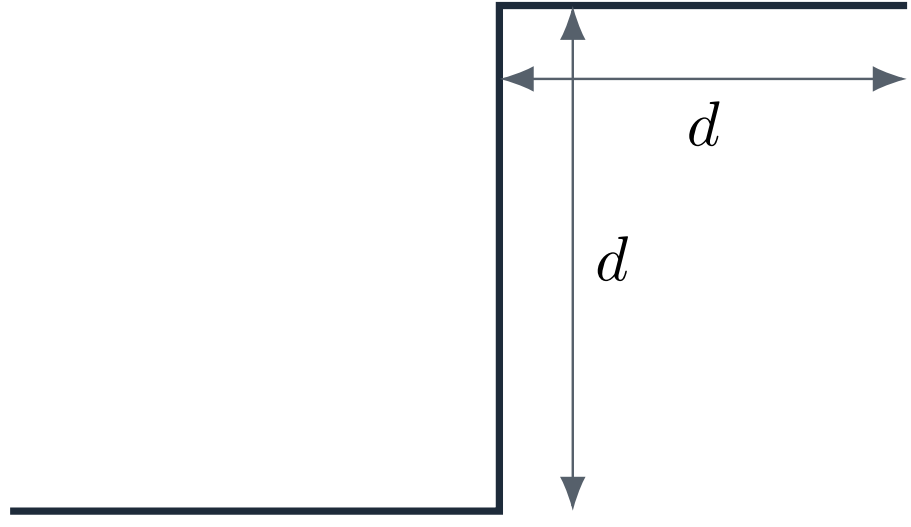
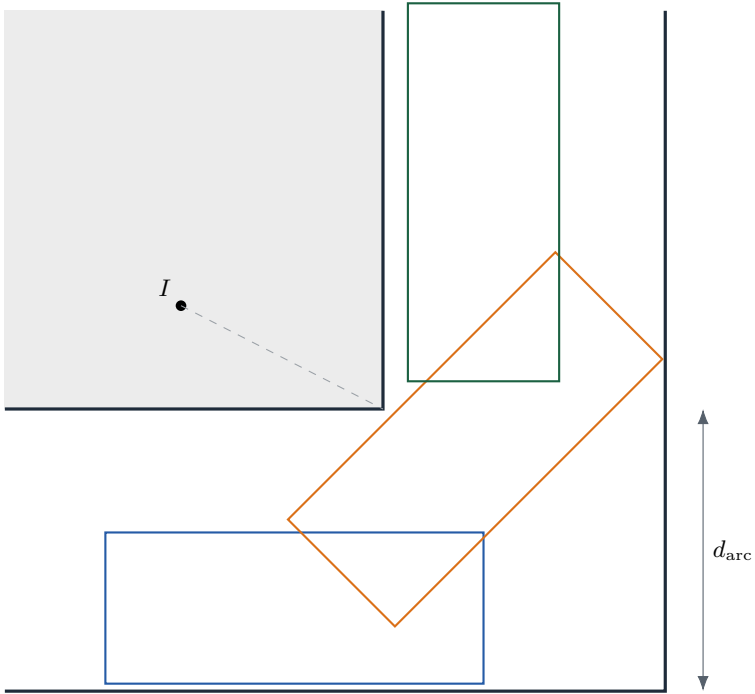

$2l$
α
β
$3l$


d
d



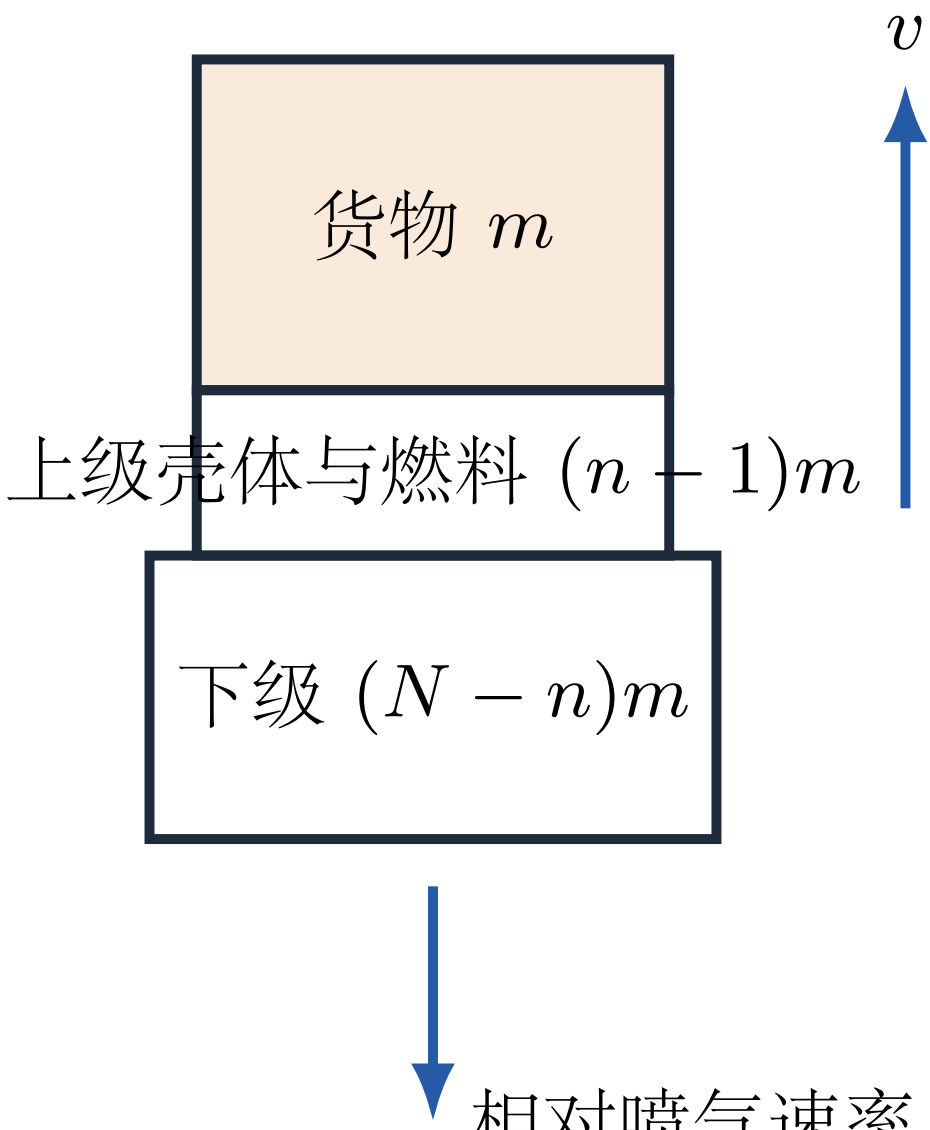
恒定最大转角的一种可行构造

$d_{\mathrm{arc}} = 3.732l$

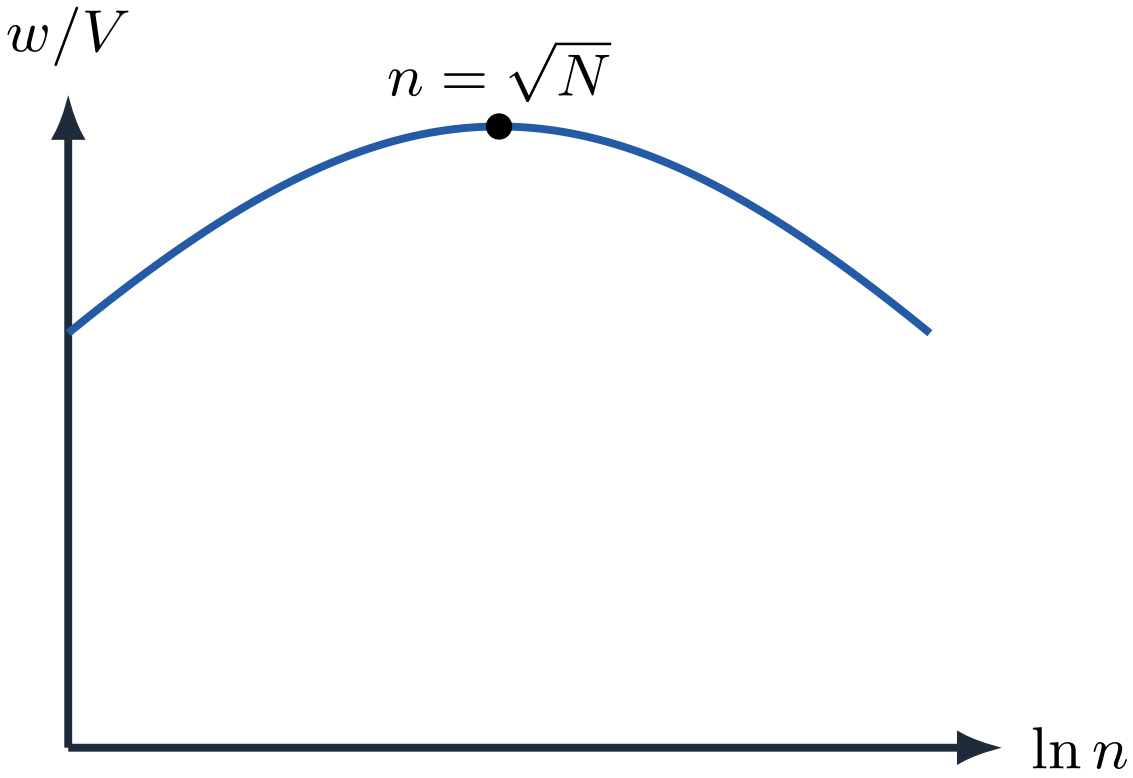
$I = (-a, d+b)$，$a^2 + b^2 = 9l^2$

灰区为道路内角外的禁区

此图不证明任意策略的全局最优



结构质量比 r，分级速度增量另记 u

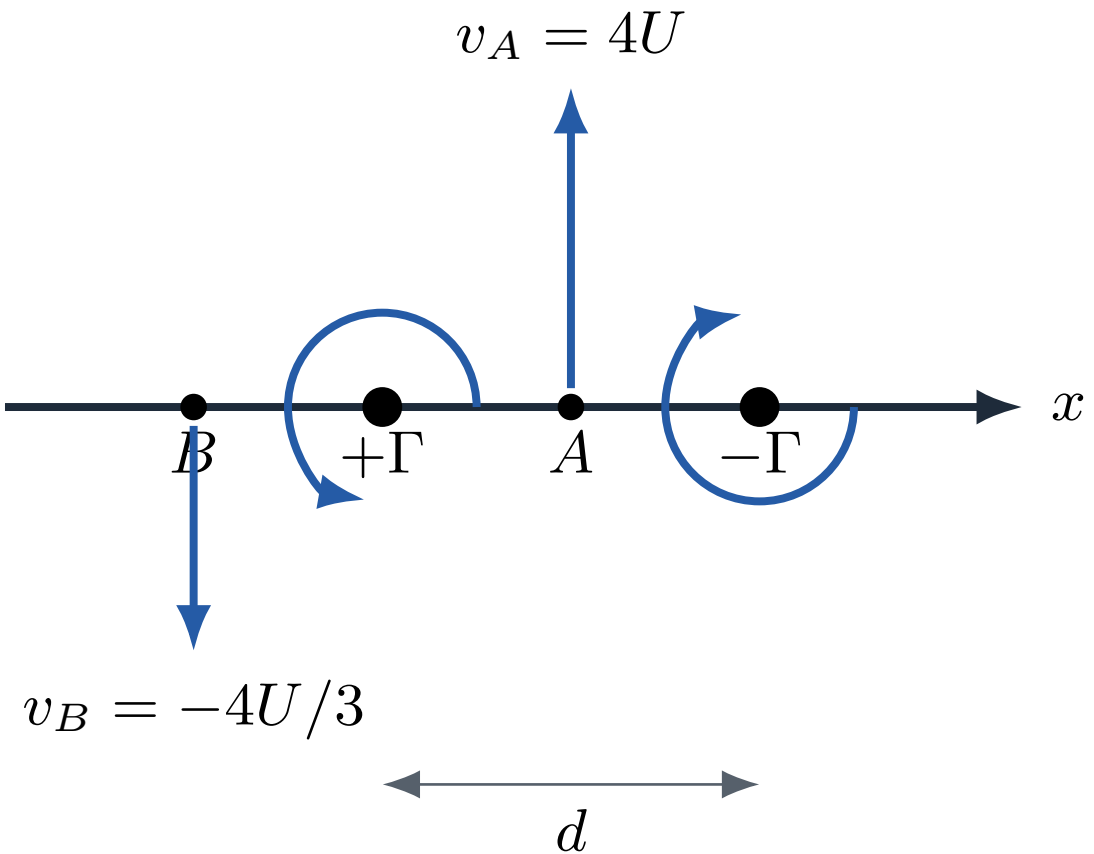


示例 $N = 100, r = 0.1$

最优点 $n = 10$

两级质量比相等

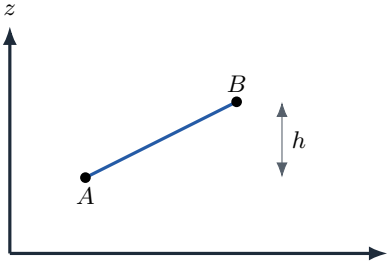
纵轴按总末速 w/V 绘制



$U = \Gamma/(2\pi d)$

自由涡对整体向上平移

箭头为实验室有向速度



位置示意

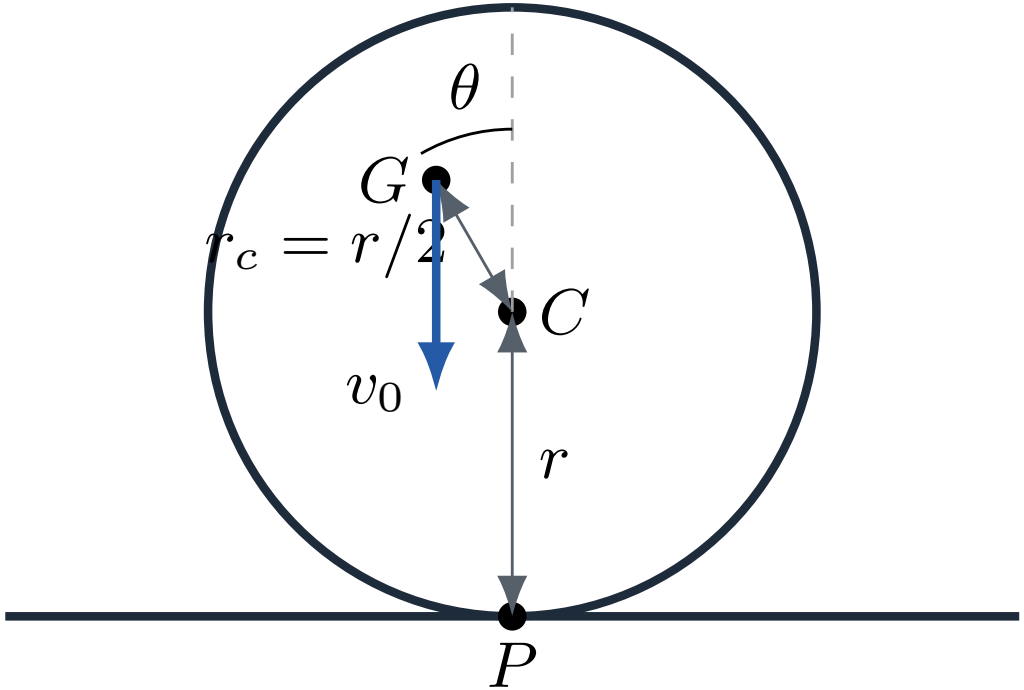
自由平移涡对：用随动速度 $3U, -7U/3$

$h = 4\Gamma^2/(9\pi^2 g d^2)$

固定涡心示意：用实验室速度 $4U, -4U/3$

$h = 16\Gamma^2/(9\pi^2 g d^2)$

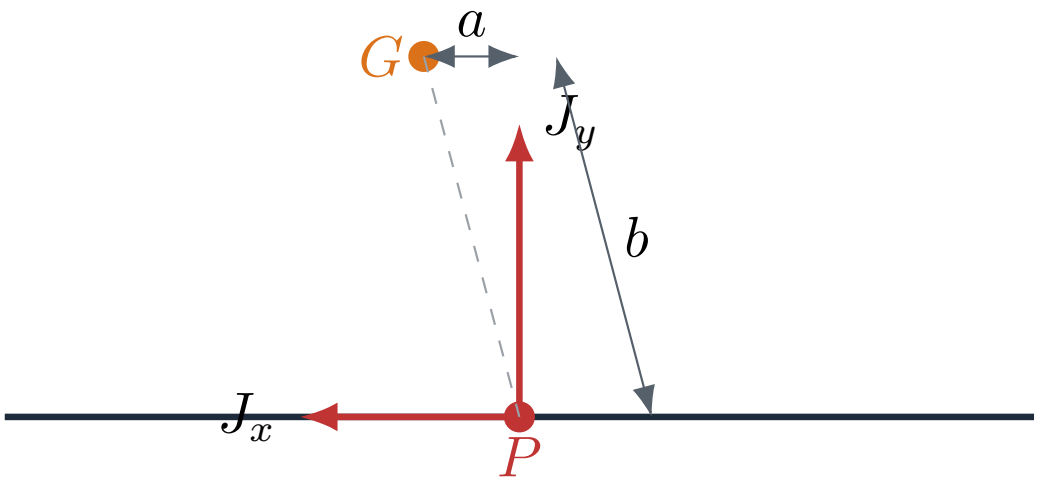
两种模型不可混用；连线不表示实际自由面形状



$r_c = r/2$

法向碰撞完全弹性

第二问另加入摩擦

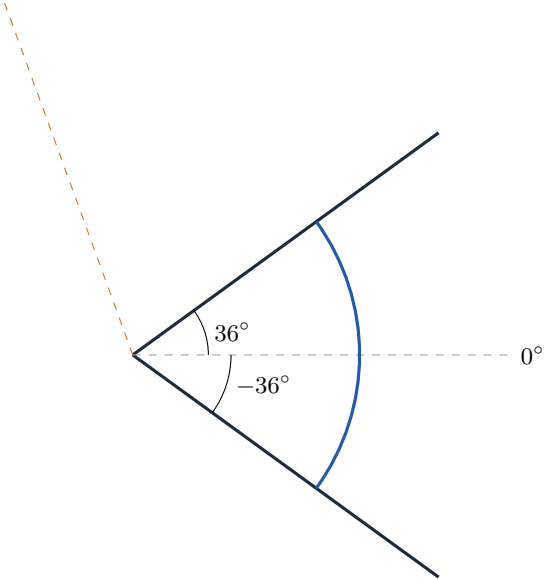


$$a = r_c \sin\theta, \quad b = r + r_c \cos\theta$$

$$v_{Py}^{+} = -v_{Py}^{-}$$

$$\mu_c = \frac{ab/r^2}{1 + b^2/r^2}$$

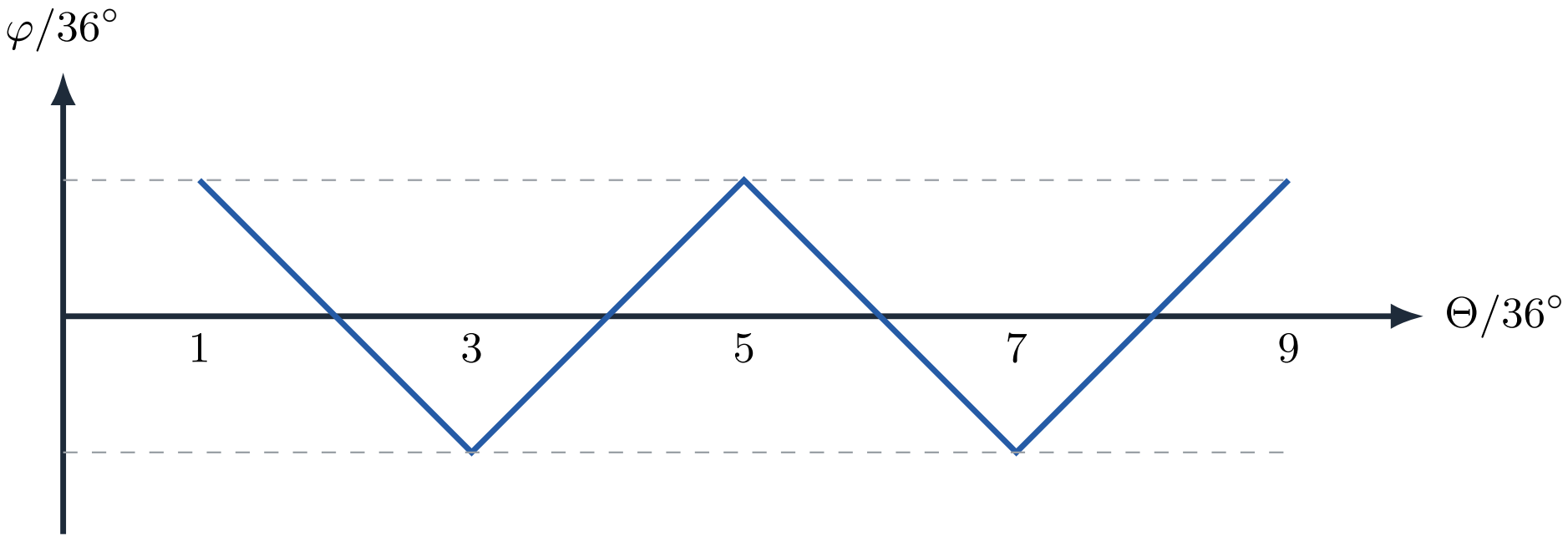
Θ：约定的绝对像角

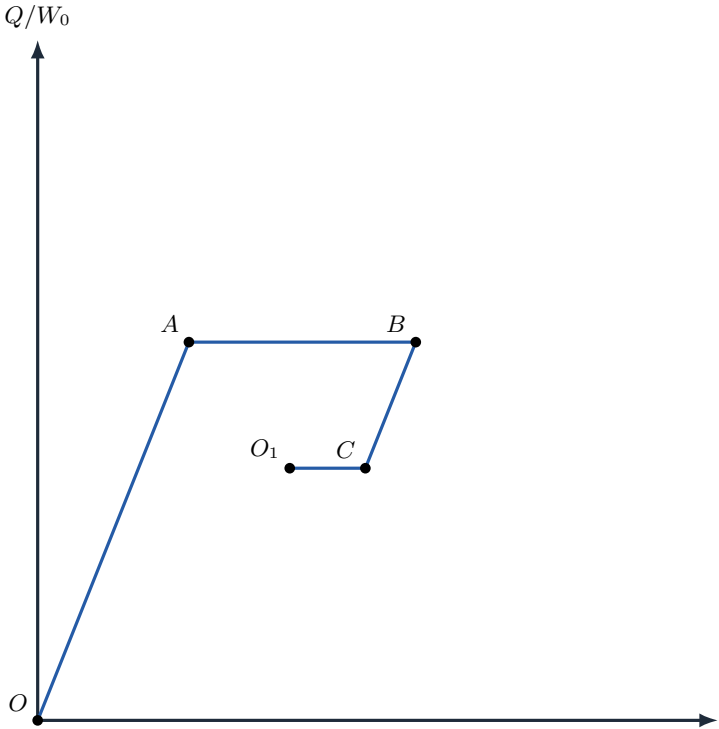


原图的观察角起点需另核实
此图只声明镜像展开的角坐标
形式像的位置不等同实际可见光路

指定展开顺序的四个分支；开区间端点不纳入

真实光路须另按观察位置与镜面有效侧筛选

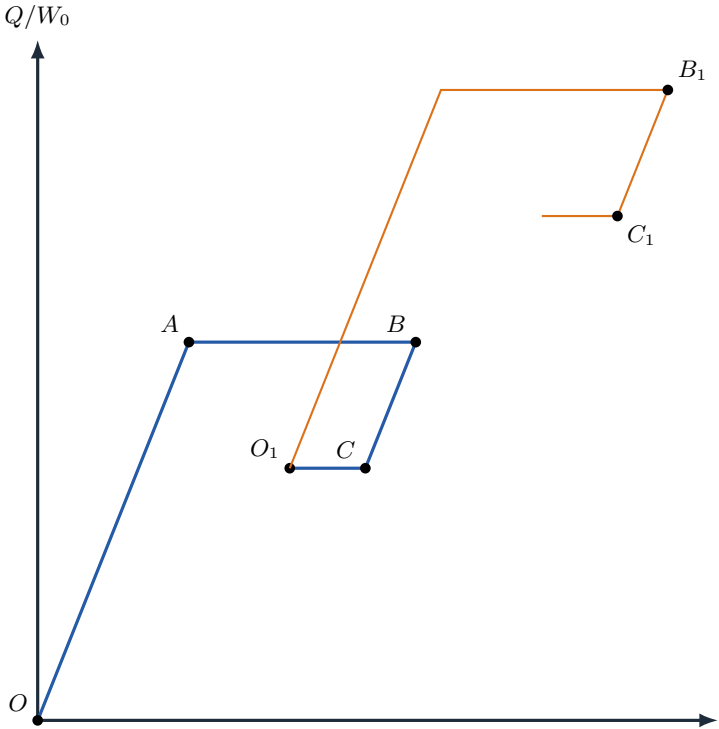




$OA \parallel BC$，$AB \parallel CO_1$

回到同一状态，但累计坐标继续增加

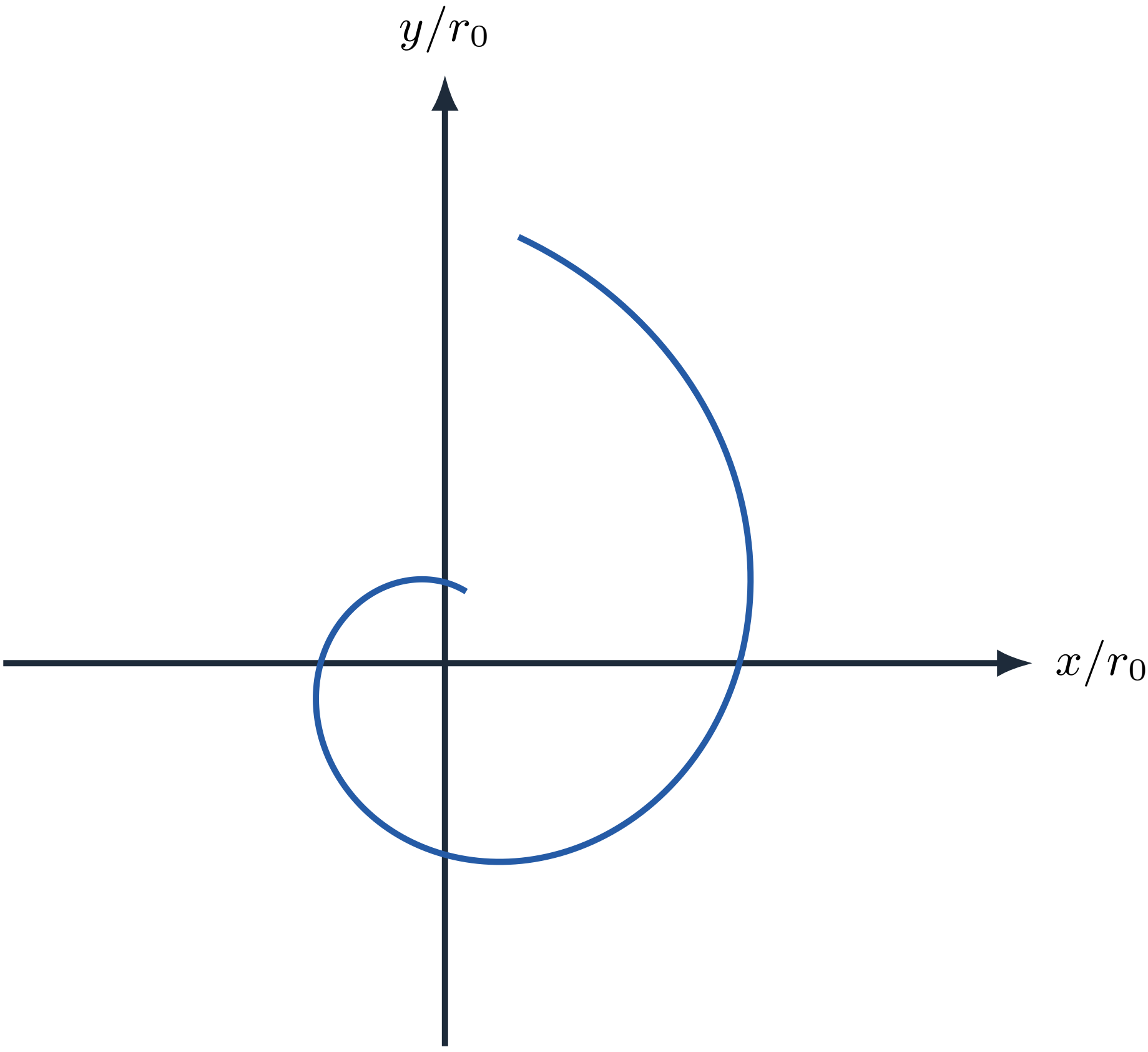
斜边对应恒定有效热容

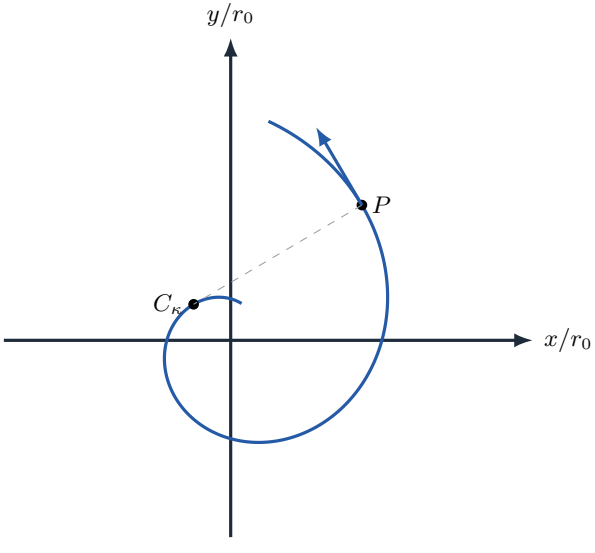


$OA \parallel BC$，$AB \parallel CO_1$

回到同一状态，但累计坐标继续增加

斜边对应恒定有效热容


y/r_0
x/r_0



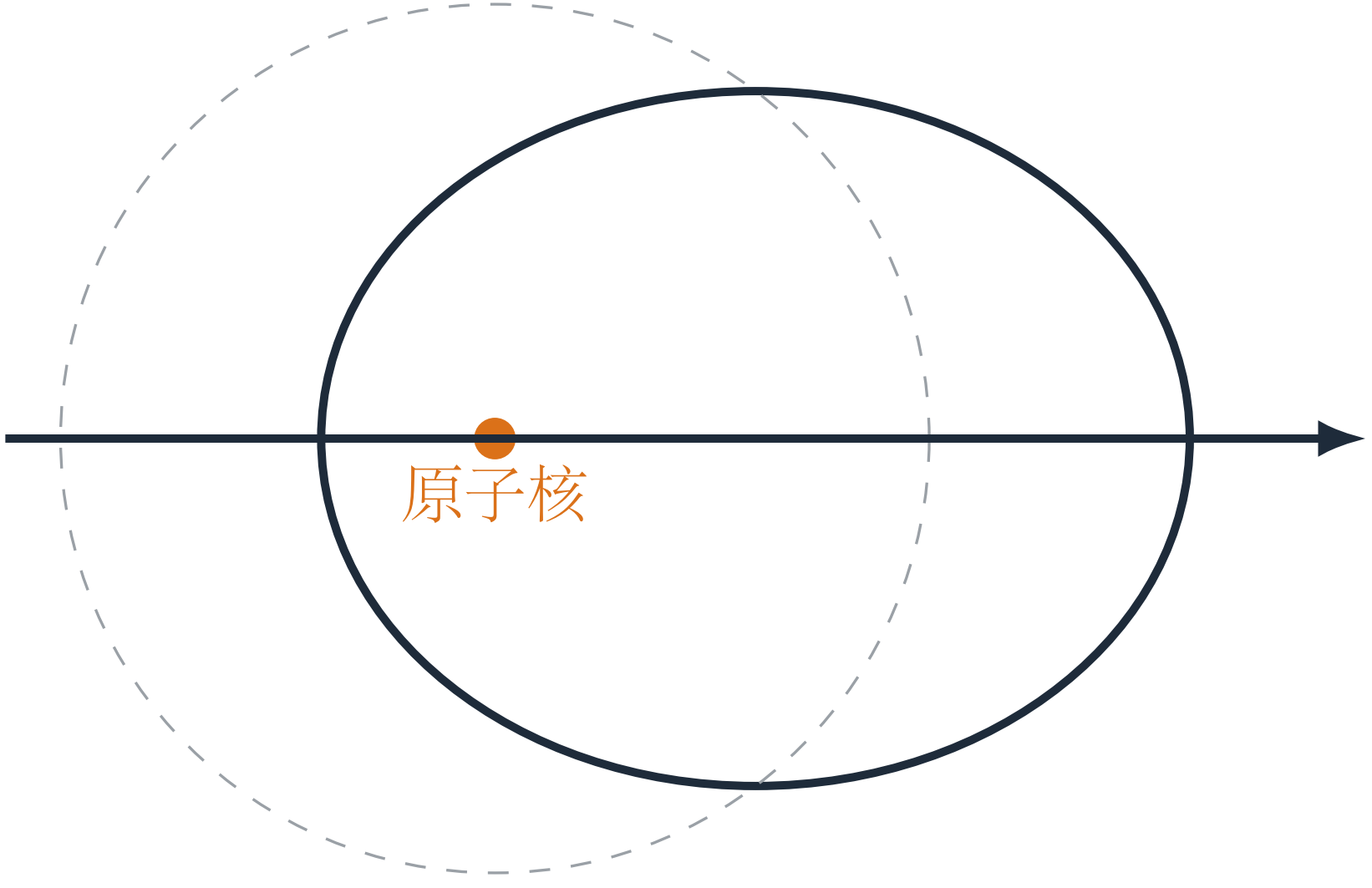
$\rho = r\sqrt{1+k^2}$

$\boldsymbol{C}_\kappa = kr\boldsymbol{e}_\theta$

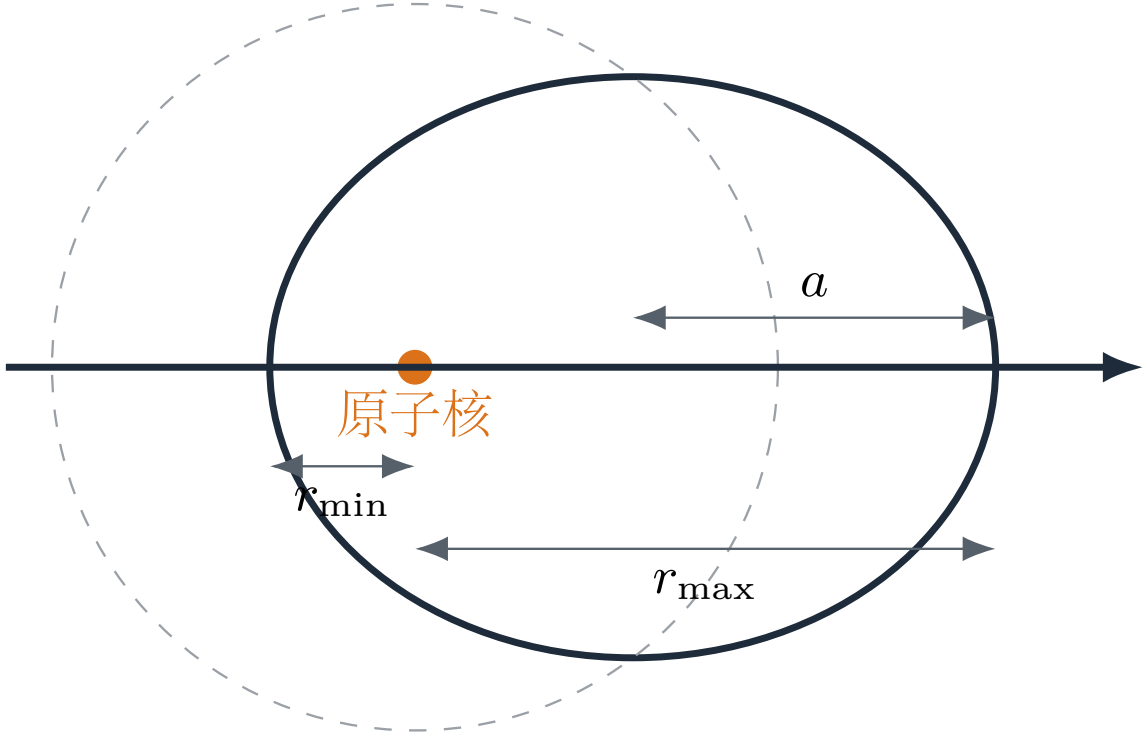
$B(r) = B_0 r_0 / r$

箭头示例取 $q > 0$、磁场入纸面

实线：椭圆；虚线：同能量圆轨道



实线：椭圆；虚线：同能量圆轨道



$a = r_n$，$e = \sqrt{1 - l^2/n^2}$

核在焦点，圆心不在核处

$r_{\min,\max} = a(1 \mp e)$